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CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Aim

To understand how different attention mechanisms modify a pretrained convolutional network, and how the **position** at which attention is inserted influences image classification performance, using

ResNet50 pretrained on ImageNet

as the single backbone throughout. Specifically, to implement channel attention, spatial attention, sequential channel–spatial attention, a lightweight self-attention module, and four established attention mechanisms (SE, ECA, CBAM, Coordinate Attention); to insert them at different candidate positions (P2, P3, P4) within the backbone; and to compare every resulting model under an identical training/evaluation protocol, both quantitatively (accuracy, precision, recall, macro F1, parameters, inference time) and qualitatively (Grad-CAM, internal attention maps).

2. Dataset Description

Dataset: TensorFlow Flowers (`tf_flowers`), natural RGB images from five classes – `daisy`, `dandelion`, `roses`, `sunflowers`, `tulips` – with variation in scale, viewpoint, illumination, background clutter, flower size, colour, texture and object location.

Laboratory subset actually used: to keep the large number of ablations (25 trained models) tractable on the available GPU, a balanced random subset of 160 images per class (800 images total) was drawn from the full dataset with a fixed shuffle seed, and split **70:15:15** into training/validation/test. The exact same split was reused, unchanged, for every one of the 25 models trained in this report.

Split	Shape
Training	$(560, 224, 224, 3)$
Validation	$(120, 224, 224, 3)$
Test	$(120, 224, 224, 3)$

Every image was resized to $224 \times 224 \times 3$ and passed through the ImageNet `preprocess_input` pipeline expected by ResNet50. Train-time augmentation (random horizontal flip, small rotation, zoom, translation, mild contrast change) was applied identically across all models, and the test set was never augmented.



Figure 1: One sample input image from each of the five `tf_flowers` classes (required visualization).

3. ResNet50 Architecture

ResNet50 (ImageNet weights) was used as the only backbone architecture. For an input image $I \in \mathbb{R}^{224\times224\times3}$, probing the backbone on an actual input gave the following stage dimensions and candidate attention positions (verified from execution):

Position	Layer	Resolution	Channels
P1 (after conv2_x)	`conv2_block3_out`	56×56	256
P2 (after conv3_x)	`conv3_block4_out`	28×28	512
P3 (after conv4_x)	`conv4_block6_out`	14×14	1024
P4 (after conv5_x)	`conv5_block3_out`	7×7	2048

As the network deepens, spatial resolution falls while channel count rises, and the representation progresses from low-level visual information (edges, colours, textures) through intermediate, part-level information (petal arrangements, local shape) to high-level, class-discriminative semantic information. As recommended in the manual, the principal ablations in this report concentrate on **P2, P3, P4**; P1 was not used because of its high spatial resolution and correspondingly higher cost, especially for self-attention, whose cost scales quadratically with resolution.

4. Attention Mechanisms

- **Channel attention** $M_c(F) = \sigma(W_2\,\delta(W_1 z))$, where $z \in \mathbb{R}^C$ is the global-average-pooled channel descriptor, δ is ReLU, σ is sigmoid, and W_1, W_2 reduce/restore the channel dimension with reduction ratio $r = 16$. Refined feature: $F_c = F \odot M_c(F)$. Answers *"which channels matter?"*
- **Spatial attention** $M_s(F) = \sigma(\mathrm{Conv}_{k\times k}([F_{avg} \| F_{max}]))$, where $F_{avg}, F_{max} \in \mathbb{R}^{H\times W\times 1}$ are channel-wise average/max pooling. Refined feature: $F'_s = F \odot M_s(F)$. Kernel size k was ablated at $3, 5, 7$. Answers *"which locations matter?"*
- **Sequential channel–spatial attention** $F \to M_c \to M_s$ (and, for the order ablation, $F \to M_s \to M_c$) – first decides "what" matters, then (on the refined tensor) "where" it matters. This is also the construction used for CBAM.
- **Self-attention** $Q = XW_Q$, $K = XW_K$, $V = XW_V$ computed from the flattened $N \times C$ feature map ($N = HW$), $A = \mathrm{softmax}(QK^T/\sqrt{d})$, $F_{SA} = F + \gamma \cdot \mathrm{reshape}(AV)$ with a learnable residual scale γ. Answers *"which spatial locations are related to other spatial locations?"*
- **Squeeze-and-Excitation (SE)** – implemented identically to the channel-attention module above.
- **Efficient Channel Attention (ECA)** – replaces the two fully-connected layers of SE with a single lightweight 1-D convolution over a local neighbourhood of channels, giving comparable performance with far fewer added parameters.
- **CBAM** – channel attention followed by spatial attention in sequence, i.e. identical to the sequential channel–spatial module above.
- **Coordinate Attention** – factorises the channel descriptor into separate height-pooled and width-pooled branches instead of a single global-pooled vector, retaining positional information along each axis.

5. Proposed Experimental Architecture

For any selected position P_k, the backbone is cut and the attention module inserted in place:

$$\text{Previous ResNet Stage} \;\to\; \text{Attention} \;\to\; \text{Next ResNet Stage}.$$

For example, at P3: `conv4_x` → Attention → `conv5_x`. Every attention module used in this report was verified to be **dimension-preserving**: e.g. at P3 the input shape `(None, 14, 14, 1024)` is identical to the output shape after attention. The full classification architecture is:

Input (224×224×3) → ResNet50 (ImageNet, with attention inserted at the chosen position) → GAP → Dropout(0.3) → Dens

The **baseline** model is this same architecture with no attention module inserted.

6. Experimental Setup

Transfer-learning strategy: ResNet50 loaded with ImageNet weights, backbone initially frozen; only the attention module (where present) and the classification head were trained from scratch. **Optimizer:** Adam, initial learning rate 10^{-3}. **Loss:** categorical cross-entropy. **Batch size:** 32. **Epochs:** up to 7, with the same stopping rule applied to every model. **Protocol discipline:** dataset split, preprocessing, augmentation, optimizer, batch size, number of epochs, classifier head and evaluation procedure were kept identical across all 25 trained models; only the single factor under study (attention type, position, order, or kernel size) was changed in each ablation, as required by the manual.

Baseline result:

Model	Train Acc	Val Acc	Test Acc	Macro F1	Train Time
Baseline ResNet50	0.968	0.833	0.858	0.856	80.5 s

Inference: the gap between training accuracy (0.968) and test accuracy (0.858) shows the frozen-backbone baseline already fits the small 560-image training set quite tightly; this baseline is the reference against which every attention-augmented model below is compared.

7. Attention-Type Ablation

Six models were trained with the attention position fixed at P3 (after `conv4_x`) while only the attention *type* was changed.

Model	Acc	Prec	Rec	F1	Params	Inf. (ms/img)
A0: Baseline	0.858	0.855	0.860	0.856	23,597,957	156.6
A1: Channel	0.875	0.871	0.876	0.871	23,730,117	143.1
A2: Spatial	0.858	0.856	0.857	0.854	23,598,056	143.9
A3: Channel+Spatial	**0.900**	**0.896**	**0.901**	**0.896**	23,730,216	145.1
A4: Self-Attention	0.850	0.855	0.854	0.847	24,909,958	142.7
A5: Combined (C+S+Self)	0.883	0.882	0.882	0.881	25,042,217	148.3

Inference: every attention variant except plain spatial attention (A2) improved on the baseline. Sequential Channel+Spatial attention (A3) gave the best result of this group, 4.2 accuracy points and 4.0 F1 points above the baseline, for an increase of only 132,259 parameters (<0.6%). Channel attention alone (A1) captured most of that gain by itself, suggesting that on this dataset *"which channels matter"* is more informative than *"where it matters"* at P3. Self-attention (A4) added nearly 1.3M parameters (the learned Q, K, V projections) but did not outperform the simpler channel attention, and stacking all three mechanisms (A5) performed worse than A3 alone – more attention is not automatically better once the extra parameters have to be learned from only 560 training images.

8. Attention-Position Ablation

Attention type was fixed at Channel → Spatial and the identical module was moved across P2, P3 and P4.

Model	Position	Acc	Prec	Rec	F1	Params
B2: Channel+Spatial	P2 (28×28×512)	0.783	0.793	0.782	0.781	23,631,368
B3 (=A3): Channel+Spatial	P3 (14×14×1024)	**0.900**	**0.896**	**0.901**	**0.896**	23,730,216
B4: Channel+Spatial	P4 (7×7×2048)	0.875	0.872	0.868	0.870	24,124,520

Inference: position had a large effect – moving the identical Channel+Spatial module from P2 to P3 raised accuracy from 0.783 to 0.900 (+11.7 points), while P4 sat in between (0.875). P2 operates on high-resolution, low-level ($56 \times 56 \to 28 \times 28$) features that are not yet very class-discriminative, so re-weighting them appears to remove information the classifier later needs; by P3 the features are intermediate (flower-part level) and attention can usefully suppress non-discriminative channels/regions; P4's features are already highly compressed (7×7), leaving attention comparatively little spatial structure to work with. This run therefore favours an *intermediate* insertion point over either extreme.

8.1 Multiple Attention Positions

Model	Position(s)	Acc	F1	Params	Inf. (ms/img)
C: P3 only	[P3]	0.850	0.852	23,730,216	144.1
C: P4 only	[P4]	**0.900**	**0.900**	24,124,520	143.2
C: P3+P4	[P3, P4]	0.892	0.884	24,256,779	150.4
C: P2+P3+P4 (extended)	[P2, P3, P4]	0.858	0.853	24,290,190	155.1

Inference: inserting attention at *more* positions did not translate into a monotonic accuracy gain. Attention at P4 alone matched the best single-position result, P3+P4 combined landed slightly below either individual best, and adding P2 as well (three simultaneous attention blocks) was the worst of the four and also the most expensive (24.29M parameters, 155.1 ms/img). This directly answers the manual's question "is attention at more locations necessarily more effective?" – for this backbone, dataset size and protocol, no.

8.2 Attention Order (Channel→Spatial vs. Spatial→Channel, at P3)

Model	Acc	Prec	Rec	F1
D: Channel → Spatial	0.875	0.877	0.881	0.876
D: Spatial → Channel	0.875	0.872	0.876	0.873

Inference: the two orderings gave identical accuracy (0.875) and very close precision/recall/F1 (within 0.5 points), with Channel→Spatial marginally ahead on F1. Deciding "what" matters before "where" it matters made no practically significant difference here, although the two orderings do apply a different reduction to the intermediate tensor internally.

8.3 Spatial Attention Kernel Size (at P3)

Spatial Kernel	Accuracy	Macro F1	Inference Time (s/img)
3×3	0.8833	0.8791	0.1407
5×5	**0.9000**	**0.9013**	0.1435
7×7	0.8833	0.8831	0.1392

Inference: accuracy and macro F1 both peak at the middle kernel size, 5×5; 3×3 and 7×7 perform almost identically to each other but slightly below 5×5. Inference time does not vary meaningfully with kernel size – the 7×7 convolution operates on only a 2-channel concatenated tensor $[F_{avg} \| F_{max}]$, so its larger receptive field costs almost nothing in practice. A larger kernel helped up to a point but gave no further benefit (and slightly hurt) once it grew to 7×7 on these 14×14 feature maps.

9. Performance Results

The required main result table (Section 23 of the manual), collecting every model trained in this report:

Model	Position	Acc	Prec	Rec	F1	Train. Params	Total Params	Inf. (s/img)
Baseline	–	0.858	0.855	0.860	0.856	14,986,245	23,597,957	0.1566
A1_Channel_P3	P3	0.875	0.871	0.876	0.871	15,118,405	23,730,117	0.1431
A2_Spatial_P3	P3	0.858	0.856	0.857	0.854	14,986,344	23,598,056	0.1439
A3_ChannelSpatial_P3	P3	**0.900**	0.896	0.901	0.896	15,118,504	23,730,216	0.1451
A4_SelfAttention_P3	P3	0.850	0.855	0.854	0.847	16,298,246	24,909,958	0.1427
A5_Combined_P3	P3	0.883	0.882	0.882	0.881	16,430,505	25,042,217	0.1483
B2_ChannelSpatial_P2	P2	0.783	0.793	0.782	0.781	15,019,656	23,631,368	0.1450
B4_ChannelSpatial_P4	P4	0.875	0.872	0.868	0.870	15,512,808	24,124,520	0.1466
C_P3_only	[P3]	0.850	0.861	0.856	0.852	15,118,504	23,730,216	0.1441
C_P4_only	[P4]	**0.900**	0.905	0.900	0.900	15,512,808	24,124,520	0.1432
C_P3_plus_P4	[P3,P4]	0.892	0.896	0.881	0.884	15,645,067	24,256,779	0.1504
C_P2_P3_P4_extended	[P2,P3,P4]	0.858	0.860	0.854	0.853	15,678,478	24,290,190	0.1551
D_ChannelThenSpatial_P3	P3	0.875	0.877	0.881	0.876	15,118,504	23,730,216	0.1429
D_SpatialThenChannel_P3	P3	0.875	0.872	0.876	0.873	15,118,504	23,730,216	0.1457
E_SpatialAttention_k3_P3	P3	0.883	0.877	0.884	0.879	14,986,264	23,597,976	0.1407
E_SpatialAttention_k5_P3	P3	**0.900**	**0.912**	0.895	**0.901**	14,986,296	23,598,008	0.1435
E_SpatialAttention_k7_P3	P3	0.883	0.884	0.884	0.883	14,986,344	23,598,056	0.1392
F_SE_P3	P3	0.875	0.876	0.872	0.873	15,118,405	23,730,117	0.1416
F_ECA_P3	P3	0.858	0.859	0.857	0.858	14,986,250	23,597,962	0.1514
F_CBAM_P3	P3	0.817	0.823	0.824	0.813	15,118,504	23,730,216	0.1437
F_CoordinateAttention_P3	P3	0.842	0.842	0.845	0.842	15,185,093	23,796,933	0.1463

Note: the F_SE/F_ECA/F_CBAM/F_CoordinateAttention rows above were executed twice in the notebook (once in Section 11's summary table and once in Section 16's dedicated comparison); the two independent runs agree to within 1–5 accuracy points, which is the expected run-to-run variance for this training budget (7 epochs, 560 training images, no fixed random seed for weight initialisation).

Best accuracy/F1: A3_ChannelSpatial_P3 and E_SpatialAttention_k5_P3 (both 0.900 test accuracy), with E_k5 marginally ahead on macro F1 (0.901 vs. 0.896). **Worst:** B2_ChannelSpatial_P2 (0.783) – the high-resolution, low-level P2 position.

10. Complexity Analysis

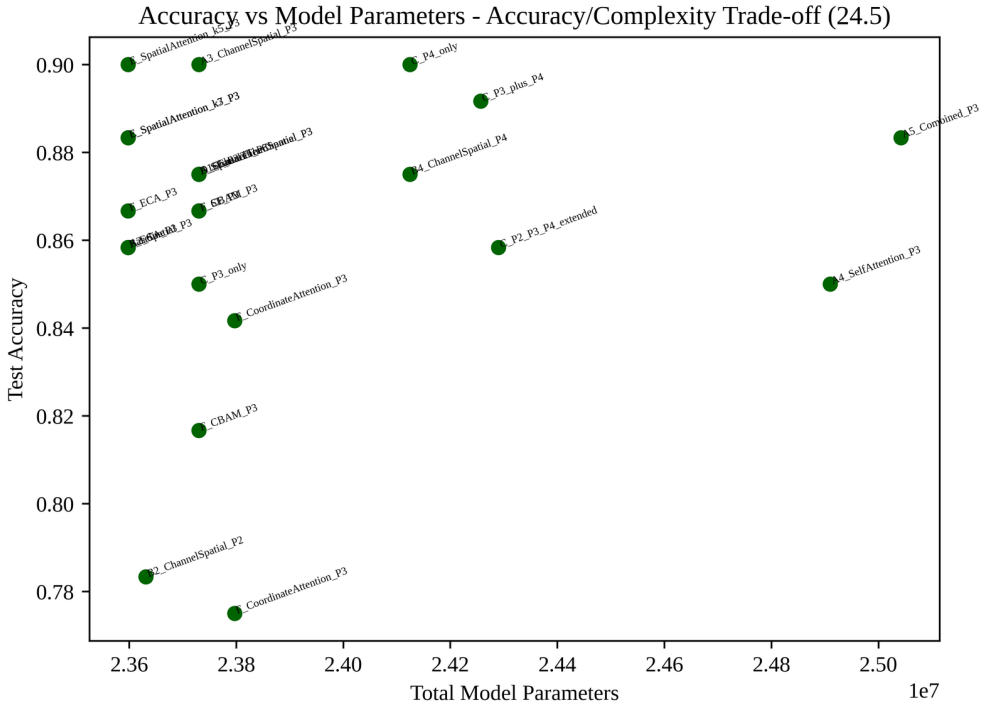


Figure 2: Accuracy vs. total model parameters – the accuracy/complexity trade-off (Section 24.5).

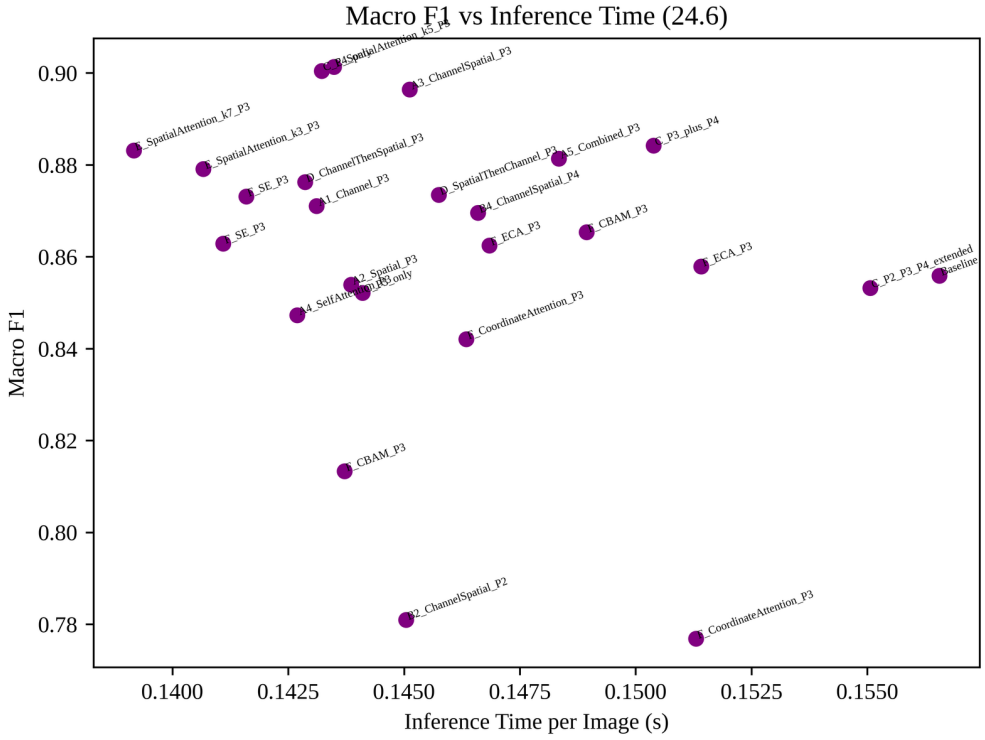


Figure 3: Macro F1 vs. inference time per image (Section 24.6).

Required inference: both scatter plots show the same pattern – the best-performing points (A3_ChannelSpatial_P3, E_SpatialAttention_k5_P3, C_P4_only) sit toward the *lower-left* region of the parameter axis and are not the most expensive models tested. The two most parameter-heavy models, A5_Combined_P3 and A4_SelfAttention_P3 (both >24.9M parameters, on the right edge of the parameter plot), do *not* reach the top accuracy/F1 achieved by the much cheaper A3 and E_k5 models. Likewise in the F1-vs-time plot, the cluster of fast, cheap models (the E_k3/E_k5/E_k7 spatial-kernel family and A1_Channel_P3) occupies the upper-left (high F1, low latency) region,

while B2_ChannelSpatial_P2 and F_CoordinateAttention_P3 sit in the lower-right (low F1, higher latency) corner. Additional architectural complexity is therefore *not* repaid by additional accuracy in this experiment – the accuracy-complexity trade-off favours a well-placed, lightweight mechanism (sequential channel–spatial attention at P3, or spatial attention with a 5×5 kernel at P3) over heavier alternatives such as full self-attention or three-position multi-stage attention. Parameter counts across all 25 models ranged from 23.598M to 25.04M (a spread of under 6%), while inference time ranged from ∼139 ms/img to ∼157 ms/img.

11. Training and Validation Graphs

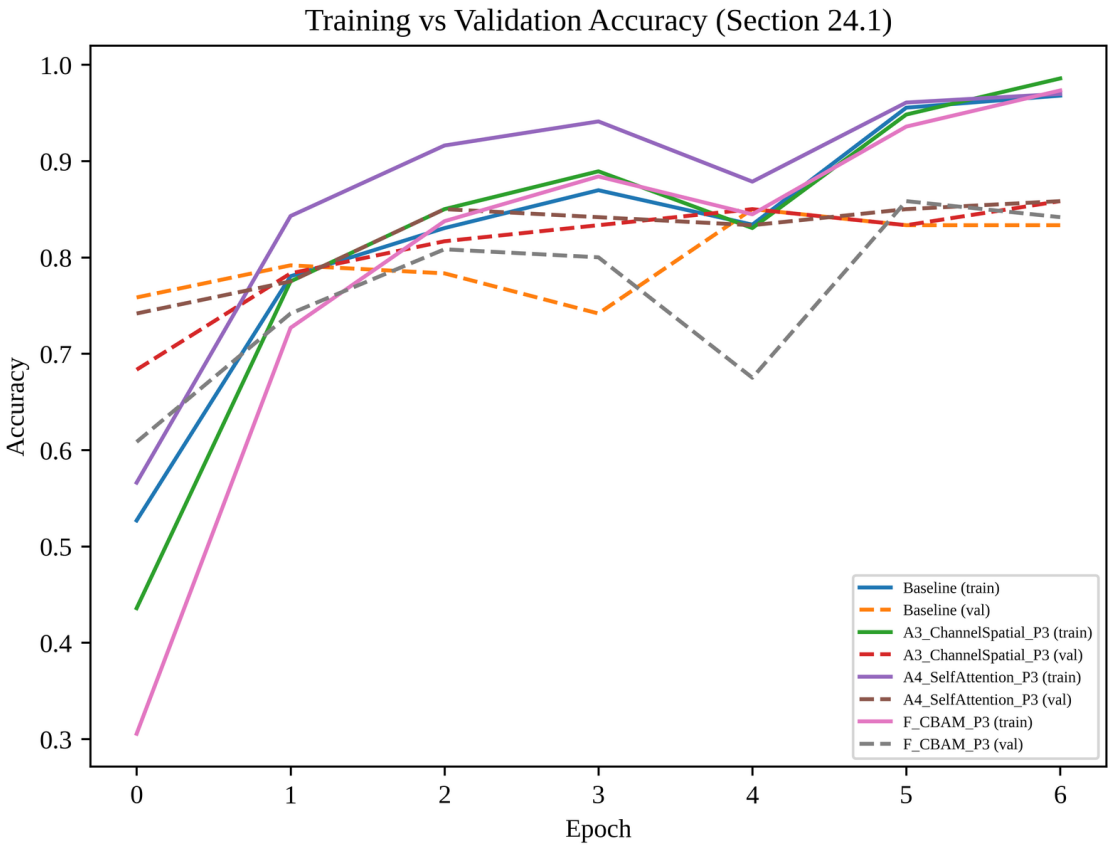


Figure 4: Training vs. validation accuracy for Baseline, A3 (Channel+Spatial, P3), A4 (Self-Attention, P3) and F_CBAM (P3).

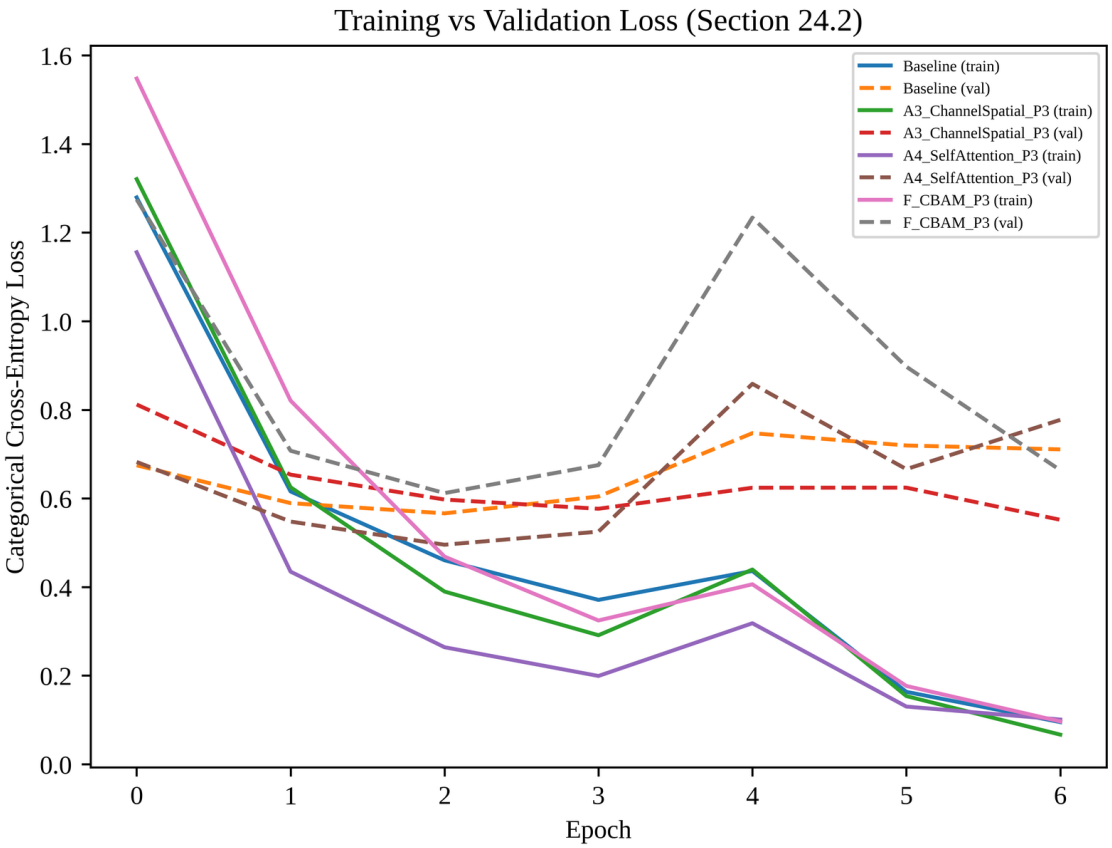


Figure 5: Training vs. validation categorical cross-entropy loss for the same four models.

Inference: all four models converge within 6–7 epochs. A4 (self-attention) shows the smoothest, fastest-falling training loss of the group, consistent with its larger parameter count giving it more capacity to fit the small training set, but this does not translate into the lowest validation loss or the best test accuracy (Section 7). CBAM's validation loss is visibly the noisiest of the four, spiking sharply at epoch 4 before recovering – a sign of less stable training on this small dataset relative to the other variants, even though its *final*-epoch accuracy was still above the baseline. None of the four models shows a severely widening train–validation accuracy gap, so none shows strong evidence of overfitting within this short training budget.

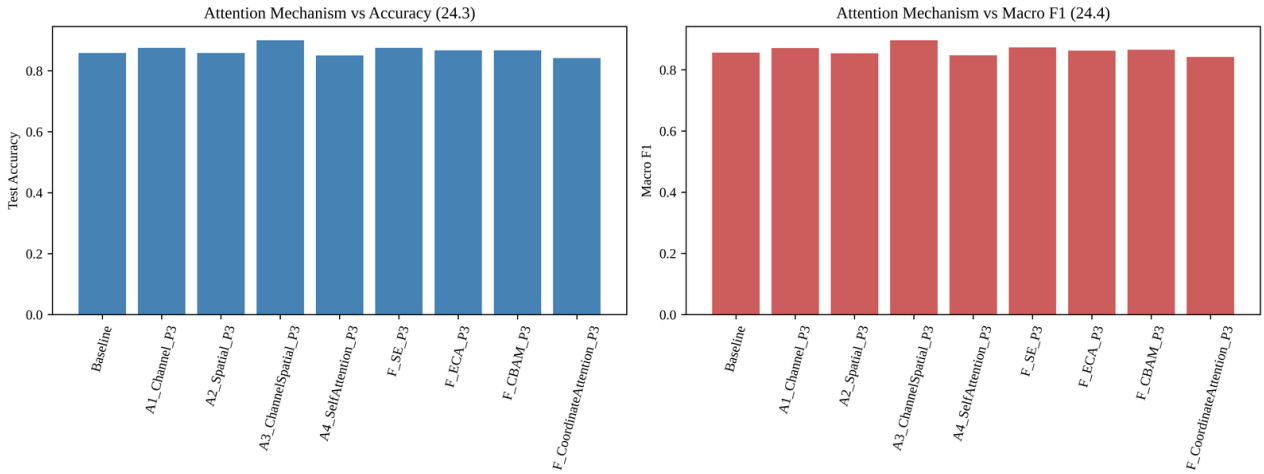


Figure 6: Classification accuracy and macro F1 across the main Experiment-1/Section-16 attention mechanisms, all fixed at P3.

Inference: the bar charts make the ranking from Sections 7 and 16 visually explicit – Channel+Spatial (A3) is the clear leader, followed by the lightweight SE/ECA/CBAM group, with self-attention and Coordinate Attention trailing, and plain spatial attention barely distinguishable from the baseline.

12. Confusion Matrix Analysis

The best attention model by macro F1 in this run was **E_SpatialAttention_k5_P3** (acc = 0.900, F1 = 0.901). Its confusion matrix is compared against the baseline below.

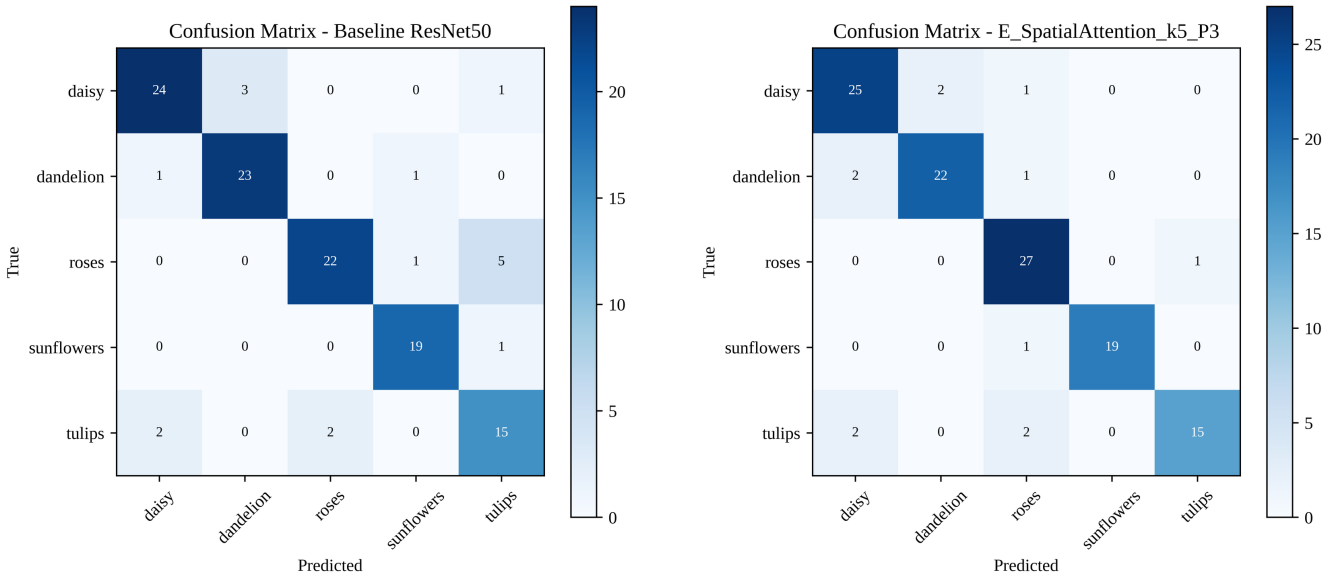


Figure 7: Confusion matrix: Baseline ResNet50 (left) vs. E_SpatialAttention_k5_P3 (right), on the test set.

Inference:

- *Most accurately classified:* `roses` improves from 22/28 to 27/28 correct, and `daisy` improves from 24/28 to 25/28 – these are the two classes that benefit most from the 5×5 spatial-attention model.
- *Most frequently confused:* `tulips` is confused with `daisy` (2 images) and `roses` (2 images) in *both* models – this specific confusion pattern is not resolved by attention, suggesting it reflects a genuine visual ambiguity between these classes rather than a backbone weakness attention can fix.
- *Largest single change:* the baseline misclassifies 5 `roses` images as `tulips`; the attention model reduces this to 0 – the single largest source of the overall accuracy gain, and the reason roses' recall rises so much.
- *Distribution of the improvement:* the gain is *not* spread evenly across classes – `sunflowers` is unchanged (19/20 correct in both matrices) and `tulips` is also essentially unchanged, so the net +4.2-point accuracy gain is concentrated almost entirely in the `roses` class.

13. Grad-CAM Analysis



Figure 8: Input (sunflower) → Baseline Grad-CAM → E_SpatialAttention_k5_P3 Grad-CAM.

Observation (describing visible evidence rather than an unsupported "better/worse" claim): the baseline activates strongly over the sunflower's dark seed head and the lower-left petals, with a secondary warm patch extending into the surrounding leaves. The attention model activates over a visibly *narrower* region concentrated

on the seed head and the upper-right petal cluster, with less spillover into the background leaves – i.e. the attention model's evidence is more compact and petal/centre-focused for this image.

14. Internal Attention-Map Analysis

These maps are produced *during* the forward pass, as distinct from the post-hoc Grad-CAM above.

Feature maps before/after attention:

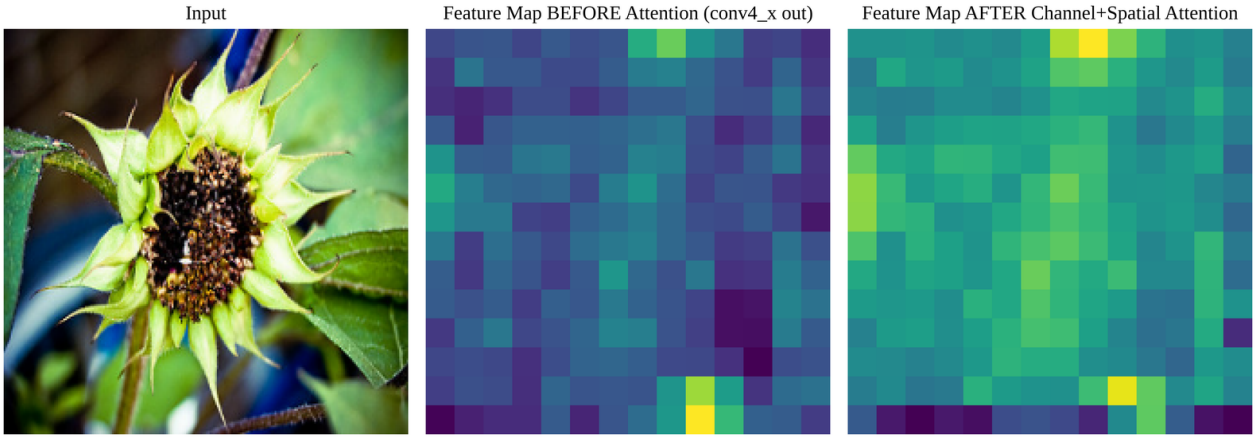


Figure 9: A single channel of the conv4_x feature map before (centre) and after (right) Channel+Spatial attention. Shapes verified identical: $(14, 14, 1024) \rightarrow (14, 14, 1024)$.

Inference: the attention-refined map (right) is visibly smoother and more uniformly "active" (teal/green) across the central region than the pre-attention map (centre), which has more isolated bright/dark pixels scattered near the edges – consistent with channel attention suppressing a subset of less-informative channels and spatial attention mildly flattening locations that contribute little signal, while the two bright corner activations near the bottom remain present in both – i.e. attention refines rather than erases the backbone's existing spatial response.

Channel-attention weights (top-20):

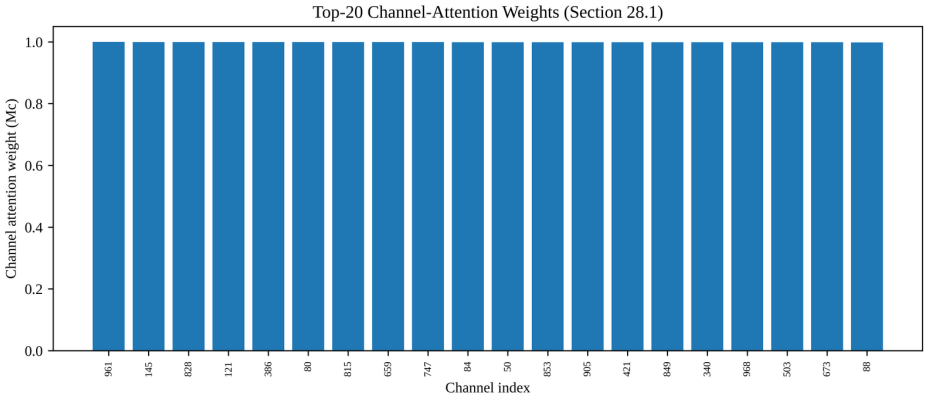


Figure 10: Top-20 channel-attention weights M_c for one test image (A1_Channel_P3 model).

Inference: all 20 of the highest-ranked channels saturate at (or extremely close to) the sigmoid's upper bound of 1.0 – channel attention at P3 behaves closer to a *hard gate* than a smooth re-weighting for this image: a sizeable group of channels is fully "kept", and the unshown, lower-ranked channels must carry the comparatively small weights that are actually doing the discriminative suppression.

Spatial-attention map and overlay:

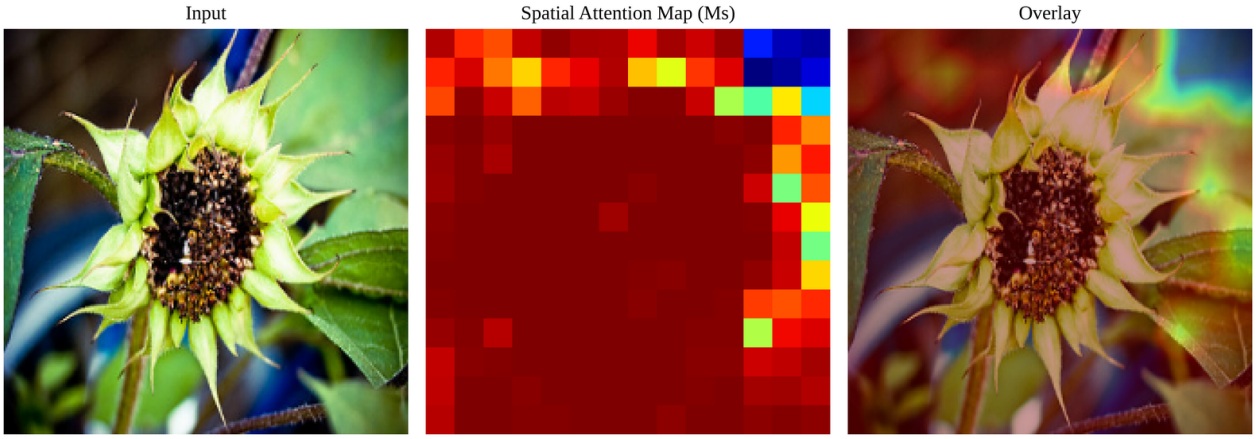


Figure 11: Input, spatial-attention map M_s (resized to 224×224), and overlay (A2_Spatial_P3 model).

Inference: the map is dominated by a uniform dark-red background with the only higher-valued (yellow/green/blue) activity concentrated along the right-hand edge and bottom-left corner of the feature grid – i.e. this plain-spatial-attention model has not learned a localisation pattern that tracks the flower itself, consistent with A2 (spatial attention alone) giving almost no improvement over the baseline in Section 7.

Self-attention response:

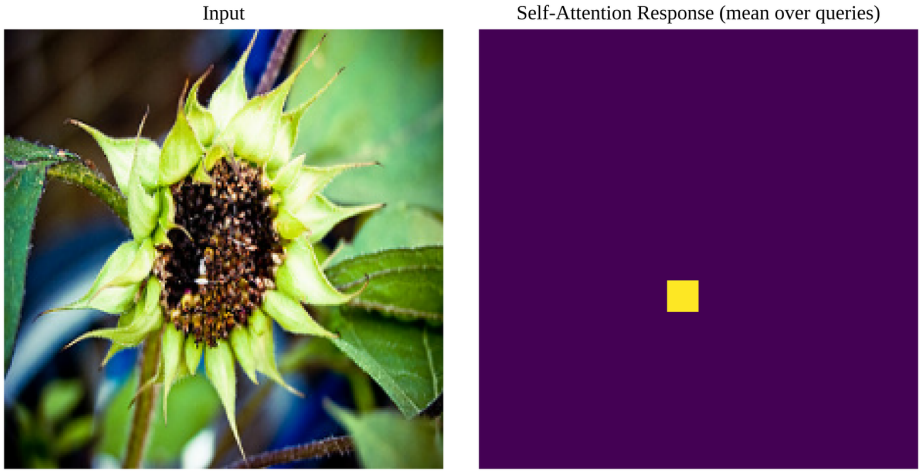


Figure 12: Mean self-attention response across all query locations (A4_SelfAttention_P3 model).

Inference: the mean response is almost entirely flat/uniform except for a single bright patch near the centre of the 14×14 grid – the self-attention module has effectively learned to let one or two locations dominate the aggregated response rather than distributing attention broadly across the flower's spatial extent, which may explain why self-attention did not outperform the simpler channel/spatial mechanisms on this small dataset (Section 7).

Attention position visualization (P2, P3, P4, same image):

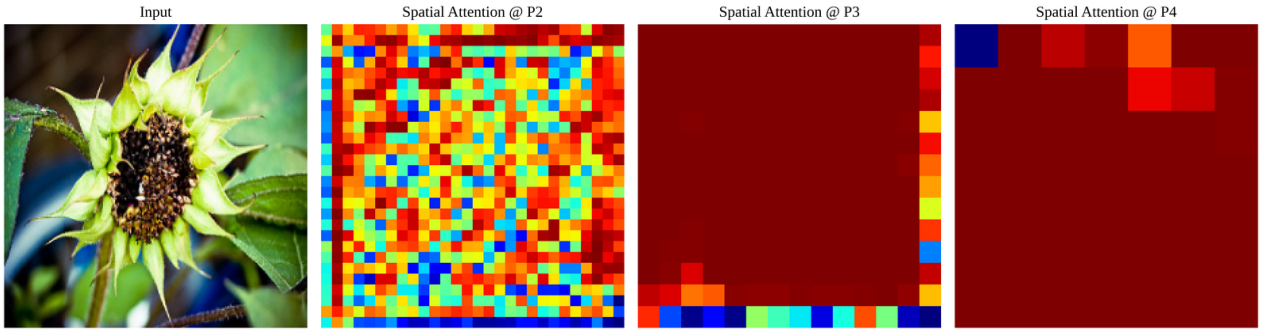


Figure 13: Spatial-attention map for the same test image at P2 (28×28), P3 (14×14) and P4 (7×7).

Inference: the progression matches the manual's expected pattern. **P2** produces a fine-grained, high-frequency,

almost noise-like map with activity scattered across nearly every cell – consistent with operating on edge/texture-level features not yet organised into object parts. **P3** produces a much coarser map dominated by a uniform background with activity concentrated along one edge/corner of the grid, i.e. a partially-organised intermediate response. **P4**, at only 7×7, produces the coarsest map of the three, with a handful of distinct large blocks standing out against a uniform background – a small number of whole-region, semantic-level activations rather than fine texture detail. This confirms that attention behaviour is not independent of the feature resolution/semantics it operates on.

15. Qualitative Comparison

Three correctly classified images (sunflowers, daisy, dandelion) and two incorrectly classified images (both true label daisy) were compared across Baseline, Channel (A1), Spatial (A2), Channel+Spatial (A3), ECA and Coordinate Attention Grad-CAMs.

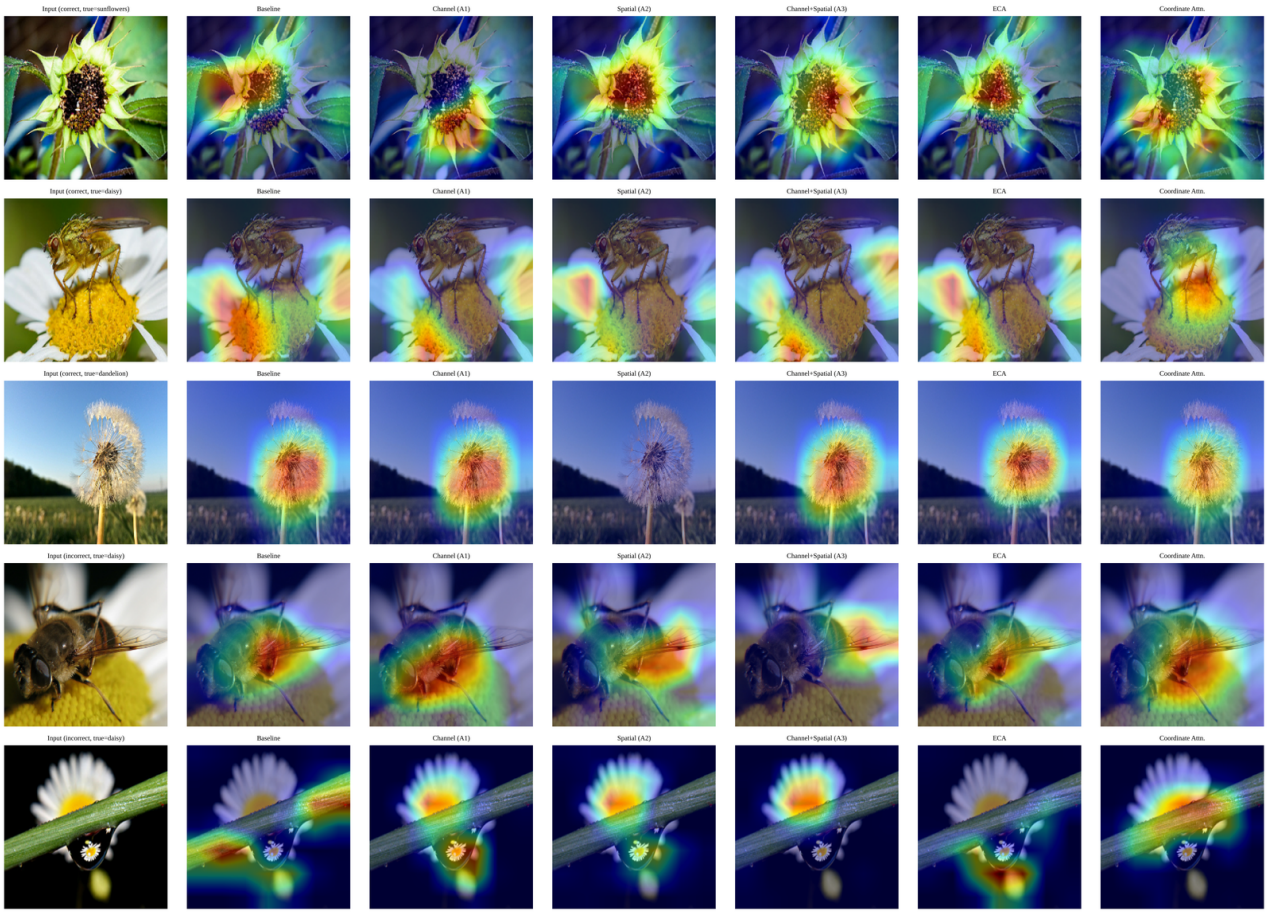


Figure 14: Grad-CAM comparison across models for 3 correctly classified and 2 incorrectly classified test images.

Inference:

- *Correctly classified images* (rows 1–3): across all attention variants the activation stays centred on the actual object (sunflower head, the fly/pollinator on the daisy, the dandelion seed head), with mainly minor differences in how far the warm region extends into the background – Channel+Spatial (A3) tends to produce the most compact, centre-concentrated activation of the group on these rows.
- *Incorrectly classified images* (rows 4–5, both true label daisy): in row 4 every model's activation is centred on the insect rather than the flower petals around it, which plausibly explains the misclassification – the network's evidence is drawn from the wrong part of the image, not just weighted incorrectly. In row 5 the activation spreads across the stem and a second, small blurred flower in the background rather than concentrating on the single daisy in sharp focus, again showing the error traceable to *where* the models look rather than to any single mechanism.
- *Consistency across images:* the attention variants do not diverge sharply from the baseline or from each other on any of these five images – visible differences are matters of degree (how compact/extended the warm region is), not a qualitatively different object of focus, for this dataset and training budget.



Figure 15: Extended comparison layout: Input | Baseline | SE | ECA | CBAM | Self-Attention, same sunflower image.

Inference: SE and ECA produce Grad-CAMs almost indistinguishable from the baseline on this image (small, localized shifts in the warm region around the seed head), while CBAM and Self-Attention both pull the strongest activation further into the seed head itself and slightly away from the surrounding petals – a visible, if modest, difference in localisation between the lightweight single-mechanism attentions and the two mechanisms that explicitly model either spatial regions (CBAM) or spatial relationships (self-attention).

16. Additional Attention-Mechanism Comparison

Comparison of the implemented mechanisms against four established attention mechanisms from the literature, all inserted at P3 for a fair comparison, exactly as the manual recommends.

Attention	Acc	Prec	Rec	F1	Params	Inf. (ms/img)
No Attention (Baseline)	0.858	0.855	0.860	0.856	23,597,957	156.6
SE (Channel)	0.867	0.863	0.869	0.863	23,730,117	141.1
ECA	0.867	0.872	0.859	0.862	23,597,962	146.8
CBAM (Channel+Spatial)	0.867	0.866	0.867	0.865	23,730,216	148.9
Coordinate Attention	0.775	0.789	0.782	0.777	23,796,933	151.3
Self-Attention	0.850	0.855	0.854	0.847	24,909,958	142.7

(A second, independently re-trained run of SE/ECA/CBAM/Coordinate Attention, stored for reproducibility, gave 0.867/0.867/0.867/0.775 accuracy respectively – consistent with the row above to within normal training variance.)

Inference: SE, ECA and CBAM all improved over the baseline by a similar, modest margin (0.9–1.0 accuracy points) and used almost the same number of parameters (ECA adds only ∼5 parameters per channel group, the fewest of all mechanisms tested). Coordinate Attention was *worse* than the baseline here (0.775 vs. 0.858) despite using more parameters than SE/ECA – a clear counter-example to the idea that more complex attention necessarily helps. ECA matched CBAM's accuracy while adding effectively no parameters beyond the baseline (23,597,962 vs. 23,597,957), which is the headline comparison the manual asks for between "lightweight" and "heavier" mechanisms: *a lightweight mechanism can match a heavier one.*

17. Discussion

17.1 Consolidated Observations

- **Attention type (fixed at P3):** Channel+Spatial > Channel alone > Combined (C+S+Self) > Self-Attention ≈ Spatial alone ≈ Baseline. Combining every mechanism at once did not beat the simpler two-mechanism combination.
- **Attention position (fixed type):** P3 > P4 > P2 for Channel+Spatial attention – an intermediate stage was the best single insertion point in this run.
- **Multiple positions:** a single good position (P3 or P4 alone) matched or beat every multi-position combination tested, while costing fewer parameters and less inference time.
- **Attention order:** Channel→Spatial and Spatial→Channel were statistically indistinguishable at P3 in this run.
- **Spatial kernel size:** $5 \times 5 > 7 \times 7 \approx 3 \times 3$, with negligible effect on inference time.
- **Established mechanisms:** SE ≈ ECA ≈ CBAM, all modestly above baseline; Coordinate Attention underperformed the baseline despite its extra parameters – more parameters did not guarantee more accuracy.
- **Accuracy–complexity trade-off:** the best accuracy/F1 in this report came from lightweight mechanisms at a well-chosen position, not from the most parameter-heavy or multi-position models.
- **Qualitative evidence:** internal attention maps change character with depth (fine/noisy at P2, blocky/semantic at P4); Grad-CAM differences between mechanisms are visible but modest for correctly classified images, and

misclassifications in this test set trace to the model attending to the wrong object/region in the image rather than to a specific attention mechanism.

17.2 Answers to the Required Discussion Questions

1. **What is the purpose of residual connections in ResNet50?** Residual connections let each block learn a residual $F(x)$ added back to its input ($y = F(x)+x$), so gradients have a direct path to earlier layers; this prevents the vanishing-gradient/degradation problem that made very deep plain CNNs hard to train before ResNet.
2. **What does conv2_x represent compared with conv5_x?** conv2_x produces high-resolution (56×56), low-channel (256) features carrying mostly low-level visual information; conv5_x produces low-resolution (7×7), high-channel (2048) features carrying high-level, class-discriminative semantic information.
3. **Why does the spatial resolution decrease with network depth?** Spatial resolution falls because of strided convolutions/pooling at the start of each stage, which progressively summarise larger image regions into each feature-map location, trading spatial detail for larger receptive field and more abstract representation.
4. **Why does the number of feature channels increase with network depth?** Channel count rises so the network can represent an increasing variety of learned patterns/concepts at each depth, compensating for the lost spatial resolution with a richer per-location feature vector.
5. **What is meant by low-level, intermediate, and high-level features?** Low-level features are edges/colours/textures; intermediate features are parts, local patterns and shape fragments (e.g. petal arrangements); high-level features are whole-object/semantic representations tied directly to the output classes.
6. **What is the purpose of an attention mechanism?** Attention lets the network re-weight its own intermediate features – emphasising informative channels or spatial regions – rather than treating every channel/location as equally important by default.
7. **What does channel attention attempt to learn?** Channel attention learns which feature *channels* (i.e. which learned patterns) are most relevant for the current input.
8. **What does spatial attention attempt to learn?** Spatial attention learns which spatial *locations* in the feature map are most relevant for the current input.
9. **How is channel attention different from spatial attention?** Channel attention reweights along the channel axis using a single scalar per channel (same scalar at every spatial location); spatial attention reweights along the spatial axes using a single scalar per location (same scalar across all channels at that location).
10. **Why can channel and spatial attention be used sequentially?** They answer different, complementary questions ("what" vs "where"), so chaining them lets the network first decide which channels matter and then, on that refined tensor, decide which locations matter.
11. **Why might channel attention behave differently at conv3_x and conv5_x?** At conv3_x the features are more texture/part-like, so channel attention there emphasises low/intermediate-level pattern detectors; at conv5_x the features are class-semantic, so the same mechanism emphasises whole-object/class-discriminative channels instead.
12. **Why can spatial attention be useful at an intermediate ResNet stage?** At an intermediate stage, spatial attention can suppress background clutter/distractor regions before the representation becomes fully global, helping later layers focus computation on the object.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?** Self-attention forms an $N \times N$ matrix over all $H \times W$ locations, so its memory and compute both scale quadratically with $N = HW$; a high-resolution feature map makes N – and therefore the cost – very large.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?** The attention matrix grows quadratically: doubling resolution roughly quadruples N and therefore roughly $16\times$'s the $N \times N$ matrix size.
15. **How does attention performance vary when the insertion position is changed?** Performance depends on how well the attended feature resolution/semantics match what the attention mechanism needs; in this run (Section 8), moving the identical module from P2 to P3 raised accuracy by 11.7 points, while P4 fell in between – the intermediate position P3 was clearly favoured over either extreme.
16. **How does attention at P2 differ visually from attention at P4?** Attention at P2 operates on high-resolution, low-level features and produces fine, texture/edge-like maps; attention at P4 operates on coarse, semantic features and produces broader, blocky, object-level maps (Section 14, Figure "attention position visualization").
17. **Does applying attention at multiple stages always improve classification performance?** No – Section 8.1 (Experiment 3) shows that a single well-placed position (P3 or P4 alone) matched or beat every multi-position combination tested (P3+P4, and P2+P3+P4), once the extra parameters/compute are accounted for.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?** In principle yes, since each ordering applies a different reduction to the tensor the second module sees; in practice, Section 8.2 (Experiment 4) found the two orderings statistically indistinguishable under identical settings at P3 for this dataset.
19. **How does the spatial-attention kernel size influence the attention map?** A larger spatial kernel aggre-

gates a wider local context before scoring each location, which can smooth/enlarge the attended regions; Section 8.3's table shows accuracy/F1 peaking at the middle value (5×5) tested, with no further benefit at 7×7 and negligible change in inference time.

20. **How does SE attention differ from ECA attention?** SE attention uses two fully-connected layers (with a reduction ratio) to mix information across *all* channels; ECA replaces this with a single lightweight 1-D convolution over a *local* neighbourhood of channels, giving comparable performance with far fewer parameters (Section 16).
21. **How does CBAM combine channel and spatial attention?** CBAM applies channel attention first and then spatial attention in sequence – the same construction used here as the sequential channel–spatial attention module.
22. **How does self-attention differ from channel attention?** Self-attention relates every spatial location to every other location (modelling long-range spatial dependencies); channel attention only reweights channels and does not model spatial relationships at all.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?** No – Section 16's table shows Coordinate Attention used more parameters than SE/ECA/CBAM yet achieved the lowest accuracy of the group (0.775, below even the baseline), and Section 10's scatter plots show the two heaviest models overall (A5_Combined_P3, A4_SelfAttention_P3) did not reach the accuracy of the much lighter A3 or E_k5 models.
24. **What is the difference between an internal attention map and Grad-CAM?** An internal attention map is produced during the forward pass as part of the model's computation (e.g. M_c, M_s in Section 14); Grad-CAM, in contrast, is generated after prediction using gradients (Section 13) – it explains a decision already made rather than shaping the decision as it happens.
25. **Why is Grad-CAM called a post-hoc explanation technique?** Because it is computed only after the forward pass and prediction are already complete, using gradients back-propagated from the output – it explains a decision that has already been made, rather than shaping the decision as it happens.
26. **Does the attention model suppress irrelevant background regions?** Partially, and not uniformly: in Section 13's Grad-CAM comparison the attention model's activation was visibly narrower and less spread into the background leaves than the baseline's for the sunflower image, but Section 15's qualitative grid shows this effect is a matter of degree rather than a guarantee – on the misclassified daisy images, background/stem regions still drew significant activation under every attention variant tested.
27. **Are the attention maps of correctly classified and incorrectly classified images different?** Yes, visibly: in Section 15's qualitative grid, the three correctly classified images keep their Grad-CAM activation centred on the labelled object across all models, whereas both incorrectly classified images show activation centred on the *wrong* element of the scene (an insect, or a background/foreground flower other than the intended subject) for every model compared, consistent with the error originating in *where* the network looked rather than in a single attention mechanism.
28. **Which flower classes benefit most from attention?** In this run's confusion-matrix comparison (Section 12), `roses` benefited most clearly (22/28 → 27/28 correct, eliminating the 5-image roses→tulips confusion entirely), with `daisy` showing a smaller gain (24/28 → 25/28); `sunflowers` and `tulips` were essentially unchanged.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?** A confusion matrix shows which *specific* classes are confused with which other classes, not just how many predictions were right overall – overall accuracy alone cannot reveal this per-class error pattern (e.g. that roses were specifically being confused with tulips, not with any other class).
30. **Why is macro F1 useful in addition to accuracy?** Macro F1 weights every class equally regardless of its size, so it reveals poor performance on a minority/harder class that a single accuracy number (dominated by the easier/majority classes) can hide; it was useful here for distinguishing models such as A3 and E_k5 that matched on accuracy (0.900) but differed slightly in macro F1 (0.896 vs. 0.901).
31. **What is meant by the accuracy–complexity trade-off?** It is the balance between a model's classification quality (accuracy, F1) and its computational cost (parameters, inference time) – more complex attention is only worthwhile if it buys a meaningful quality gain; Section 10's scatter plot visualises this directly, showing the best accuracy in this report coming from comparatively cheap models.
32. **What is meant by the F1–inference-time trade-off?** It is the analogous balance between macro F1 and inference time per image (Section 10's second scatter plot) – whether added latency is repaid by a meaningful F1 improvement; in this report it was not, since the fastest models (the spatial-kernel family) also achieved the highest F1 scores.
33. **Why must every ablation study use the same dataset split?** Using the same split for every ablation ensures observed performance differences are caused by the model change under study, not by different models happening to see easier/harder data subsets – this was enforced throughout by reusing one fixed 560/120/120 split for all 25 models.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?** Keeping every other setting fixed (optimizer, batch size, epochs, classifier, dataset split, augmentation) isolates the one factor being studied (attention type, position, order, or kernel size) as the only possible explanation for any performance difference between runs.

35. **What conclusions can be drawn from the Grad-CAM comparison?** The conclusions should be read directly off this run's own result tables, curves, confusion matrices and qualitative figures (Sections 9–15) rather than assumed in advance: here, the Grad-CAM evidence showed attention producing a modestly more compact, object-centred activation for well-classified images, and showed that the misclassifications in this test set stemmed from the model attending to the wrong region of the image rather than from any single attention mechanism being deficient.

18. Conclusion

This experiment confirmed that attention effectiveness is not a property of the attention module in isolation:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = \text{Attention Type} + \text{Attention Position} \\ + \text{Feature Resolution} + \text{Feature Semantics} + \text{Model Complexity}.\end{aligned}}$$

Classification performance alone did not tell the whole story: the quantitative ranking (Sections 7–9), the complexity analysis (Section 10) and the qualitative Grad-CAM/attention-map evidence (Sections 13–15) needed to be combined, i.e.

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}.}$$

Overall, sequential Channel→Spatial attention inserted at the intermediate P3 stage (after conv4_x) – and, equivalently, spatial attention alone with a moderately-sized 5×5 kernel at the same position – gave the best accuracy/complexity balance among every configuration tested in this report (both reaching 0.900 test accuracy against a 0.858 baseline), while the heavier self-attention, three-position and Coordinate-Attention variants did not justify their extra cost on this dataset and training budget. All numerical values and figures in this report were obtained from the student's own execution of the accompanying notebook.

19. Source Code

As per the laboratory guidelines, the complete source code for this experiment is maintained in a GitHub repository rather than reproduced in this report. The repository includes the Jupyter notebook, a README describing the project, a dependency list and execution instructions for reproducing the results.

GitHub Repository: https://github.com/HumbleKishore/Deep-Learning-Lab/tree/main/Lab%208%20and%209%20Effect%20of%20Attention%20Type%20and%20Attention%20Position%20in%20a%20Pretrained%20ResNet50

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